



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

February/March 2026

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

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Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are mandatory marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication

Question	Answer	Marks
1(a)	angle (subtended at the centre of a circle) when arc (length) = radius	B1
1(b)(i)	$v = r\omega$	C1
	radius = $2.4 / 0.75$ = 3.2 m	A1
1(b)(ii)	$\omega = 2\pi f$	C1
	frequency = $0.75 / 2\pi$ = 0.12 Hz	A1
1(b)(iii)	$a = r\omega^2$ = $(3.2 \times \frac{2}{3}) \times 0.75^2$	C1
	= 1.2 m s⁻²	A1
1(c)	linear velocity magnitude: X greater than Z	B1
	linear velocity direction: opposite	B1
	angular velocity: magnitude equal <u>and</u> direction same	B1

Question	Answer	Marks
2(a)	force per unit positive charge	B1
2(b)(i)	four parallel lines perpendicular to the plates, each starting on one plate and finishing on the other	B1
	four parallel lines, approximately evenly distributed over the region between the plates	B1
	arrows showing direction as upwards	B1

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Question	Answer	Marks
2(b)(ii)	gravitational force is downwards so electric force must be upwards	B1
	electric force is upwards so Q must be positive	B1
2(c)(i)	$E = V / x$	A1
2(c)(ii)	$W = VQ / x$	A1
2(d)	$W = mg$ <u>and</u> $g = 9.81$	C1
	mass = $(4800 \times 3 \times 1.60 \times 10^{-19}) / (0.024 \times 9.81)$ = $9.8 \times 10^{-15} \text{ kg}$	A1
2(e)	both (electric and gravitational) forces are now downwards	B1
	droplet will <u>accelerate</u> downwards	B1

Question	Answer	Marks
3(a)	thermal energy per unit mass (to cause change of state)	B1
	(thermal) energy required to cause change of state between solid and liquid without change in temperature	B1
3(b)	<u>change</u> in volume is greater for vaporisation	B1
	<i>either:</i> increase in volume so work is done against the atmosphere <i>or:</i> work done against atmosphere so more thermal energy supplied	B1

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Question	Answer	Marks
3(c)	$Q = mc\Delta\theta$	C1
	evidence of correct application of conservation of thermal energy [e.g. loss in energy of water as it cools = gain in energy of original ice as it warms + gain in energy of ice as it melts]	C1
	$[(34 - \theta) \times 132.4 \times 4.19] = [(\theta \times 15.6 \times 4.19) + (18 \times 15.6 \times 2.11) + (15.6 \times 336)]$	C1
	temperature = 21 °C	A1

Question	Answer	Marks
4(a)	ΔU : <u>increase</u> in internal energy (of the system)	B1
	Q: thermal energy transferred <u>to</u> the system	B1
4(b)(i)	$W = p\Delta V$	C1
	$= (2.0 \times 10^5) \times (0.048 - 0.012) = 7200 \text{ J}$	A1
4(b)(ii)	<i>either:</i> volume decreases so W is positive <i>or:</i> gas is compressed so W is positive	B1
4(b)(iii)	<i>either:</i> volume is proportional to (thermodynamic) temperature and (thermodynamic) temperature is proportional to internal energy	B1
	volume quarters so internal energy quarters	B1
	<i>or:</i> volume quarters so (thermodynamic) temperature quarters	(B1)
	(thermodynamic) temperature quarters so internal energy quarters	(B1)

Question	Answer	Marks
4(c)	$\Delta U = 3.6 - 14.4$ $= -10.8 \text{ kJ}$	C1
	$Q = -10.8 - 7.2$ $= -18.0 \text{ kJ}$	A1

Question	Answer	Marks
5(a)	number of particles per unit amount of substance	B1
5(b)	<u>any three points from:</u> - the gas is ideal - volume (of gas) = 0.46 m³ <u>up to two points from:</u> <i>either:</i> mass of one molecule (of gas) = $6.6 \times 10^{-27} \text{ kg}$ <i>or:</i> mass of one mole (of gas) = $4.0 \times 10^{-3} \text{ kg}$ <i>either:</i> amount of gas = 2.6 mol <i>or:</i> number of molecules (of gas) = 1.6×10^{24} <i>either:</i> mass of the gas = 0.010 kg <i>or:</i> density of the gas = 0.022 kg m^{-3}	B3
5(c)(i)	<i>either:</i> gas is ideal so yes <i>or:</i> gas obeys $pV \propto T$ so yes <i>or:</i> gas obeys $\langle c^2 \rangle \propto T$ so yes <i>or:</i> gas obeys $E_K \propto T$ so yes	B1

Question	Answer	Marks
5(c)(ii)	<u>any two points from:</u> - molecules are in <u>constant</u> random motion (all) collisions (involving molecules) are elastic - no intermolecular forces (except during collisions) (all) collisions (involving molecules) are instantaneous	B2

Question	Answer	Marks
6(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): <i>either:</i> acceleration is (always) in the opposite <u>direction</u> to displacement <i>or:</i> acceleration is (always) <u>directed</u> towards a fixed point	B1
6(b)(i)	$\omega = 2\pi / T$ and ω identified as 6.9	C1
	$T = 2\pi / 6.9$ $= \mathbf{0.91\ s}$	A1
6(b)(ii)	$v_0 = \omega x_0$	C1
	$= 6.9 \times 3.2 = 22\ \text{mm s}^{-1}$	A1
6(c)(i)	$E = BA / t$	C1
	$E_0 = 0.047 \times 2 \times 0.060 \times (22 \times 10^{-3} \Delta t) / \Delta t$	C1
	$= \mathbf{1.2 \times 10^{-4}\ V}$	A1
6(c)(ii)	<u>sketch:</u> two cycles of a sinusoidal curve of period T and amplitude E_0	B1
	sinusoidal curve of correct phase, with $E = 0$ at (and only at) $t = 0.25T$, $t = 0.75T$, $t = 1.25T$ and $t = 1.75T$	B1

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Question	Answer	Marks
7(a)	the constant / direct current (in a resistor)	M1
	that causes the same <u>mean</u> power dissipation as the alternating current	A1
7(b)(i)	mean power = 3.2 W	A1
7(b)(ii)	$P = I^2 R$	C1
	$I_{\text{RMS}} = \sqrt{(3.2 / 26)} \text{ or } [\sqrt{(6.4 / 26)}] / \sqrt{2}$ = 0.35 A	A1
7(c)(i)	diode	B1
7(c)(ii)	<u>sketch:</u> <i>either:</i> line showing $P = 0$ between $t = 0$ and $t = 0.0125 \text{ s}$ and between $t = 0.0375 \text{ s}$ and $t = 0.050 \text{ s}$ and non-zero P in between	M1
	<i>or:</i> line showing $P = 0$ between $t = 0.0125 \text{ s}$ and $t = 0.0375 \text{ s}$, and non-zero P either side	
	line in non-zero P regions with correct $\sin^2$ shape, sitting on the t -axis, with maximum P at 6.4 W	A1
7(c)(iii)	mean power is now half of what it was before rectification	B1
	$I_{\text{RMS}} = \sqrt{(1.6 / 26)}$ = 0.25 A	A1

Question	Answer	Marks
8(a)	<u>any three points from:</u> • <i>either:</i> photons contain discrete amounts of energy <i>or:</i> energy of photons depends on wavelength (of radiation) • photon energy equals difference between energy levels • discrete wavelengths (of spectral lines) means electron transitions must involve discrete energy changes • discrete energy changes means energy levels must be discrete	B3
8(b)(i)	galaxy moving away (from Earth) causes redshift	B1
	<u>observed</u> wavelength is greater (than emitted wavelength)	B1
8(b)(ii)	four lines drawn, each to the right of the corresponding line in Fig. 8.1	B1
	approximately equal separation of each drawn line from its corresponding original line	B1
8(b)(iii)	$\nu = H_0 d$ $= 2.1 \times 10^{-18} \times 6.7 \times 10^{24}$ $= 1.4(1) \times 10^7 \text{ m s}^{-1}$	C1
	$\Delta\lambda = (1.41 \times 10^7 \times 656) / (3.0 \times 10^8)$ $= 31 \text{ nm}$	C1
	wavelength = 656 + 31 = 687 nm	A1

Question	Answer	Marks
9(a)	spontaneous emission of (ionising) radiation	B1
	emission (of radiation) from an unstable nucleus	B1
9(b)(i)	$\lambda = \ln 2 / (12.0 \times 60)$	C1
	$= 9.63 \times 10^{-4} \text{ s}^{-1}$	A1
9(b)(ii)	$A = (5.80 \times 10^{14}) \exp (-\ln 2 \times 20.0 / 12.0)$	C1
	$= 1.83 \times 10^{14} \text{ Bq}$	A1
9(b)(iii)	<u>sketch:</u> line starting at (0, 5.80)	B1
	line with negative gradient from $t = 0$ to $t = 20.0$ min with gradient of continuously decreasing magnitude	B1
	line passing through (12, 2.90) <u>and</u> (20, 1.83)	B1
9(c)	<u>any three points from:</u> - radiation is emitted in all directions but only some is towards detector - some radiation is absorbed within the sample - some radiation is absorbed in the air (before reaching detector) - some radiation is absorbed by the window of the detector - some radiation reaches the detector but does not register a count	B3

Question	Answer	Marks
10(a)(i)	ultrasound waves make the <u>crystal</u> in the transducer vibrate	B1
	vibrations (of crystal) cause an <u>induced</u> e.m.f. (that is detected)	B1
10(a)(ii)	product of density and speed	M1
	speed of sound in medium (and density of medium)	A1

Question	Answer	Marks
10(b)(i)	speed = $(1.49 \times 10^6) / 998$	C1
	= 1490 m s⁻¹	A1
10(b)(ii)	coefficient = $[(9.98 - 1.49) / (9.98 + 1.49)]^2$	C1
	= 0.548	A1
10(b)(iii)	<u>any three points from:</u> • (specific acoustic) impedance values are <u>very</u> different • intensity reflection coefficient is (approximately) equal to 1 • (almost) all the sound (intensity) is reflected (at the boundary) • (virtually) none of the sound (intensity) is transmitted (into the glass)	B3